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Communication Engineering Department
Applied Electromagnetics
Worksheet 1-Solution

1. Three field quantities are given by:

$$\mathbf{P} = 2 \mathbf{a}_x - \mathbf{a}_z$$

$$\mathbf{Q} = 2 \mathbf{a}_x - \mathbf{a}_y + 2 \mathbf{a}_z$$

$$\mathbf{R} = 2 \mathbf{a}_x - 3 \mathbf{a}_y + \mathbf{a}_z$$

Determine the followings:

- a. $(\mathbf{P} + \mathbf{Q}) \times (\mathbf{P} - \mathbf{Q})$.
- b. $\mathbf{Q} \cdot \mathbf{R} \times \mathbf{P}$
- c. $\mathbf{P} \cdot \mathbf{Q} \times \mathbf{R}$
- d. $\sin \theta_{QR}$
- e. $\mathbf{P} \times (\mathbf{Q} \times \mathbf{R})$
- f. A unit vector perpendicular to both $\mathbf{Q}$ and $\mathbf{R}$
- g. The component of $\mathbf{P}$ along $\mathbf{Q}$

Solution

a.

$$(\mathbf{P} + \mathbf{Q}) \times (\mathbf{P} - \mathbf{Q}) = \mathbf{P} \times (\mathbf{P} - \mathbf{Q}) + \mathbf{Q} \times (\mathbf{P} - \mathbf{Q}) = \mathbf{P} \times \mathbf{P} - \mathbf{P} \times \mathbf{Q} + \mathbf{Q} \times \mathbf{P} - \mathbf{Q} \times \mathbf{Q}$$

$$= \mathbf{0} + \mathbf{Q} \times \mathbf{P} + \mathbf{Q} \times \mathbf{P} - \mathbf{0} = 2\mathbf{Q} \times \mathbf{P}$$

$$= 2 \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 2 & -1 & 2 \\ 2 & 0 & -1 \end{vmatrix}$$

$$= 2(1 - 0)\mathbf{a}_x + 2(4 + 2)\mathbf{a}_y + 2(0 + 2)\mathbf{a}_z = 2 \mathbf{a}_x + 12 \mathbf{a}_y + 4 \mathbf{a}_z$$

b. The only way $\mathbf{Q} \cdot \mathbf{R} \times \mathbf{P}$ makes sense is:

$$\mathbf{Q} \cdot (\mathbf{R} \times \mathbf{P}) = (2, -1, 2) \cdot \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 2 & -1 & 2 \\ 2 & 0 & -1 \end{vmatrix}$$

$$= (2, -1, 2) \cdot (3, 4, 6) = 6 - 4 + 12 = 14$$

Alternatively:

$$\mathbf{Q} \cdot (\mathbf{R} \times \mathbf{P}) = \begin{vmatrix} 2 & -1 & 2 \\ 2 & -3 & 1 \\ 2 & 0 & -1 \end{vmatrix}$$

To find the determinant of a 3×3 matrix, we repeat the first two rows and cross multiply; when the cross multiplication is from right to left, the result should be negated as shown below. This technique of finding a determinant applies only to a 3×3 matrix. Hence:

$$\begin{aligned} \mathbf{Q} \cdot (\mathbf{R} \times \mathbf{P}) &= \begin{vmatrix} 2 & -1 & 2 \\ 2 & -3 & 1 \\ 2 & 0 & -1 \\ 2 & -1 & 2 \\ 2 & -3 & 1 \end{vmatrix} \\ &= +6 + 0 - 2 + 12 - 0 - 2 = 14 \end{aligned}$$

c. From the equation:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

Then:

$$\mathbf{P} \cdot (\mathbf{Q} \times \mathbf{R}) = \mathbf{Q} \cdot (\mathbf{R} \times \mathbf{P}) = 14$$

or

$$\begin{aligned} \mathbf{P} \cdot (\mathbf{Q} \times \mathbf{R}) &= (2, 0, -1) \cdot (5, 2, -4) \\ &= 10 + 0 + 4 = 14 \end{aligned}$$

d.

$$\begin{aligned} \sin \theta_{QR} &= \frac{|\mathbf{Q} \times \mathbf{R}|}{|\mathbf{Q}||\mathbf{R}|} = \frac{|(5, 2, -4)|}{|(2, -1, 2)||(-2, -3, 1)|} \\ &= \frac{\sqrt{45}}{3\sqrt{14}} = \frac{\sqrt{5}}{\sqrt{14}} = 0.5976 \end{aligned}$$

e.

$$\mathbf{P} \times (\mathbf{Q} \times \mathbf{R}) = (2, 0, -1) \times (5, 2, -4) = (2, 3, 4)$$

Alternatively, using the bac-cab rule:

$$\begin{aligned} \mathbf{P} \times (\mathbf{Q} \times \mathbf{R}) &= \mathbf{Q}(\mathbf{P} \cdot \mathbf{R}) - \mathbf{R}(\mathbf{P} \cdot \mathbf{Q}) \\ &= (2, -4, 2)(4 + 0 - 1) - (2, -3, 1)(4 + 0 - 2) = (2, 3, 4) \end{aligned}$$

f. A unit vector perpendicular to both $\mathbf{Q}$ and $\mathbf{R}$ is given by:

g.

$$\mathbf{a} = \frac{\pm \mathbf{Q} \times \mathbf{R}}{|\mathbf{Q} \times \mathbf{R}|} = \frac{\pm(5, 2, -4)}{\sqrt{45}} = \pm(0.745, 0.298, -0.596)$$

Note that $|\mathbf{a}| = 1$, $\mathbf{a} \cdot \mathbf{Q} = 0 = \mathbf{a} \cdot \mathbf{R}$. Any of these can be used to check $\mathbf{a}$.

h. The component of $\mathbf{P}$ along $\mathbf{Q}$ is:

$$\begin{aligned} \mathbf{P}_Q &= |\mathbf{P}| \cos \theta_{PQ} \mathbf{a}_Q \\ &= (\mathbf{P} \cdot \mathbf{a}_Q) \mathbf{a}_Q = \frac{(\mathbf{P} \cdot \mathbf{Q}) \mathbf{Q}}{|\mathbf{Q}|^2} \\ &= \frac{(4 + 0 - 2)(2, -1, 2)}{(4 + 1 + 4)} = \frac{2}{9} (2, -1, 2) \\ &= 0.4444 \mathbf{a}_x - 0.2222 \mathbf{a}_y + 0.4444 \mathbf{a}_z \end{aligned}$$

2. $\mathbf{E}$ and $\mathbf{F}$ are vector fields given by $\mathbf{E} = 2x \mathbf{a}_x + \mathbf{a}_y + yz \mathbf{a}_z$ and $\mathbf{F} = xy \mathbf{a}_x - y^2 \mathbf{a}_y + xyz \mathbf{a}_z$. Determine:

- $|\mathbf{E}|$ at (1, 2, 3).
- The component of $\mathbf{E}$ along $\mathbf{F}$ at (1, 2, 3).
- A vector perpendicular to both $\mathbf{E}$ and $\mathbf{F}$ at (0, 1, -3) whose magnitude is unity.

Solution

- At (1,2,3), $\mathbf{E} = (2, 1, 6)$

$$|\mathbf{E}| = \sqrt{4 + 1 + 36} = \sqrt{41} = 6.403$$

- At (1,2,3), $\mathbf{F} = (2, -4, 6)$

$$\begin{aligned}\mathbf{E}_F &= (\mathbf{E} \cdot \mathbf{a}_F) \mathbf{a}_F = \frac{(\mathbf{E} \cdot \mathbf{F}) \mathbf{F}}{|\mathbf{F}|^2} = \frac{36}{56} (2, -4, 6) \\ &= 1.286 \mathbf{a}_x - 2.571 \mathbf{a}_y + 3.857 \mathbf{a}_z\end{aligned}$$

- At (0,1, -3), $\mathbf{E} = (0, 1, -3)$, $\mathbf{F} = (0, -1, 0)$

$$\mathbf{E} \times \mathbf{F} = \begin{vmatrix} 0 & 1 & -3 \\ 0 & -1 & 0 \end{vmatrix} = (-3, 0, 0)$$

$$\mathbf{a}_{E \times F} = \frac{\pm \mathbf{E} \times \mathbf{F}}{|\mathbf{E} \times \mathbf{F}|} = \pm \mathbf{a}_x$$

- If $\mathbf{A} = -\mathbf{a}_x + 6 \mathbf{a}_y + 5 \mathbf{a}_z$ and $\mathbf{B} = \mathbf{a}_x + 2 \mathbf{a}_y + 3 \mathbf{a}_z$. Then, find:

- The scalar projections of $\mathbf{A}$ on $\mathbf{B}$.
- The vector projection of $\mathbf{B}$ on $\mathbf{A}$.
- The unit vector perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$.

Solution

- The scalar projections of $\mathbf{A}$ on $\mathbf{B}$:

$$\mathbf{A}_B = \mathbf{A} \mathbf{a}_B = \frac{\mathbf{A} \mathbf{B}}{|\mathbf{B}|} = \frac{-1 + 12 + 15}{\sqrt{1 + 4 + 9}} = \frac{26}{\sqrt{14}} = 6.95$$

- The vector projection of $\mathbf{B}$ on $\mathbf{A}$:

$$\begin{aligned}\mathbf{B}_A &= (\mathbf{B} \cdot \mathbf{a}_A) \mathbf{a}_A = \frac{(\mathbf{B} \cdot \mathbf{A}) \mathbf{A}}{|\mathbf{A}|^2} = \frac{26}{(1 + 36 + 25)} (-1, 6, 5) \\ &= -0.4193 \mathbf{a}_x + 2.516 \mathbf{a}_y + 2.097 \mathbf{a}_z\end{aligned}$$

- The unit vector perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} -1 & 6 & 5 \\ 1 & 2 & 3 \end{vmatrix} = 8 \mathbf{a}_x + 8 \mathbf{a}_y - 8 \mathbf{a}_z$$

$$\begin{aligned}\mathbf{a}_{A \times B} &= \frac{8 \mathbf{a}_x + 8 \mathbf{a}_y - 8 \mathbf{a}_z}{8\sqrt{1 + 1 + 1}} = \frac{\mathbf{a}_x + \mathbf{a}_y - \mathbf{a}_z}{\sqrt{3}} \\ &= 0.577 \mathbf{a}_x + 0.577 \mathbf{a}_y - 0.577 \mathbf{a}_z\end{aligned}$$